

Explicit Orchestrated Decision Design (EODD):

A Design Science Method for Human-
Centred, Interpretable AI Decision Systems

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Table of Contents

Table of Contents	2
1. Introduction	5
1.1 Research Questions	5
1.2 Hypotheses	6
1.3 Contributions	6
2. Literature Review	7
2.1 Explainable AI and Interpretability	7
2.2 Multi-Agent Systems and LLM Orchestration	7
2.3 Human-Centred AI and Oversight	8
2.4 Cognitive Load and AI Interaction.....	8
2.5 Design Science Research in AI.....	9
3. Methodology	9
3.1 Design Cycle: EODD Method Development	9
Step 1: Decision Context Definition	9
Step 2: Decision Decomposition	10
Step 3: Specialised LLM Role Assignment	10
Step 4: Orchestration Rules Specification	10
Step 5: Decision Synthesis	10
Step 6: Decision Trace Generation	10
Step 7: Human Review Integration.....	11
3.2 Evaluation Methodology.....	11
Phase 1: Heuristic Development (N=5 Human Experts).....	11
3.3 Data Analysis.....	12
4. Results.....	12
4.2 Comparative User Study Results	13
4.2.1 Hypothesis 1: Interpretability	14
4.2.2 Hypothesis 2: Trustworthiness	14
4.2.3 Hypothesis 3: Oversight Performance	14
4.2.4 System Preference	14
4.2.5 Trace Quality Analysis.....	14
4.2.6 Mental Demand Analysis	15
4.3 Qualitative Findings	15
4.4 Triangulation Analysis.....	16
5. Discussion	16
5.1 Theoretical Implications	16

5.2 Practical Implications	17
5.3 Limitations and Future Research	18
6. Conclusion	19
References.....	20
Appendix A: EODD Method Detailed Specification	21
A.1 Specialised Agent Roles.....	21
A.2 Orchestration Rule Types	22
A.3 Decision Trace Structure	22
A.4 Mental Demand Measurement	23
A.5 EODD Studio Platform	23

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Abstract

As large language models (LLMs) are increasingly deployed in high-stakes decision-making contexts, the opacity of monolithic AI systems poses significant risks to safety, accountability, and human oversight. This paper presents Explicit Orchestrated Decision Design (EODD), a novel design science method that structures AI decision systems as orchestrated multi-agent architectures with explicit, human-auditable decision traces. Through a rigorous Design Science Research (DSR) methodology, we developed a seven-step design method comprising: (1) Decision Context definition, (2) Decision Decomposition, (3) Specialized LLM Role assignment, (4) Orchestration Rules specification, (5) Decision Synthesis, (6) Decision Trace generation, and (7) Human Review integration. The method was evaluated through a three-phase empirical validation process: Phase 1 involved Heuristic Development with specialized human experts ($N = 5$); Phase 2 conducted a Scaled Feasibility Assessment utilizing an innovative LLMs-as-Experts framework ($N = 100$); and Phase 3 executed Empirical User Validation via a within-subjects comparative study with graduate-level peers, teaching assistants, and domain novices ($N = 16$). Results demonstrate that EODD significantly outperforms monolithic baselines on interpretability (Cohen's $d = 1.61$, $p < .001$), trustworthiness ($d = 1.47$, $p < .001$), and error-spotting confidence ($d = 1.18$, $p < .01$). Notably, these improvements were achieved without significantly increasing mental demand ($p = .37$), indicating that enhanced interpretability does not impose additional cognitive burden on users. Expert evaluation confirmed engineering feasibility ($M = 5.66/7$) and domain-agnostic applicability (47% reported no significant friction). These findings establish EODD as a practical, generalizable method for designing interpretable, human-centred AI decision systems across diverse high-stakes domains.

The EODD Studio web application (<https://eodd.studio>) demonstrates the practical implementation of this method, providing a platform for designing, deploying, and evaluating orchestrated AI decision systems with built-in traceability and human oversight capabilities.

Keywords: Design Science Research, Human-Centred AI, Explainable AI, Multi-Agent Systems, LLM Orchestration, Decision Support Systems, AI Governance, Cognitive Load, Interpretable Machine Learning

1. Introduction

The rapid advancement of large language models (LLMs) has catalysed their deployment across high-stakes domains including healthcare diagnostics, financial risk assessment, legal reasoning, and autonomous systems (Bommasani et al., 2021; Weidinger et al., 2022). While these systems demonstrate remarkable capabilities in complex reasoning tasks, their monolithic architectures—where a single model generates decisions through opaque internal processes—present fundamental challenges for interpretability, accountability, and human oversight (Rudin, 2019; Lipton, 2018). When AI systems make errors in consequential contexts, the inability to understand how decisions were reached impedes human intervention, undermines trust, and complicates regulatory compliance (European Commission, 2021).

Contemporary approaches to AI interpretability have primarily focused on post-hoc explanations (Ribeiro et al., 2016) or attention-based visualisations (Vaswani et al., 2017), which provide approximations of model behaviour rather than faithful accounts of decision processes. These methods, while valuable, do not address the fundamental architectural opacity of monolithic systems. Furthermore, the emerging paradigm of LLM-based multi-agent systems (Wang et al., 2024; Xi et al., 2023) offers potential for decomposed reasoning but lacks systematic design methods that ensure interpretability, traceability, and human oversight by construction.

This research addresses this critical gap by developing Explicit Orchestrated Decision Design (EODD), a Design Science Research (DSR) method for creating human-centred, interpretable AI decision systems. EODD structures AI decisions as orchestrated multi-agent processes where specialised LLM agents perform distinct reasoning functions, explicit orchestration rules govern their interactions, and comprehensive decision traces enable human inspection and intervention. The method embodies the principle that interpretability should be architected into systems by design, not retrofitted through post-hoc explanations. The practical instantiation of this method is available through the EODD Studio web platform (<https://eodd.studio>), which demonstrates how the design principles can be operationalised in a deployable system.

1.1 Research Questions

This study addresses three interrelated research questions derived from the DSR framework:

1. **RQ1 (Design):** How should an orchestrated LLM design method be structured to enable interpretable, trustworthy, and human-oversight-capable AI decision systems?

2. **RQ2 (Evaluation):** Does the EODD method produce AI systems that demonstrate superior interpretability, trustworthiness, and human oversight performance compared to monolithic baselines?
3. **RQ3 (Generalisability):** Can the EODD method be applied across diverse high-stakes domains without significant domain-specific friction or increased cognitive burden?

1.2 Hypotheses

Based on the theoretical foundations of decomposed reasoning and multi-agent orchestration, we formulate three hypotheses for empirical testing:

1. **H1 (Interpretability):** EODD-based systems will demonstrate significantly higher interpretability ratings compared to monolithic baseline systems.
2. **H2 (Trustworthiness):** EODD-based systems will demonstrate significantly higher trustworthiness ratings compared to monolithic baseline systems.
3. **H3 (Oversight Performance):** Users will demonstrate significantly higher confidence in identifying errors within EODD-based systems compared to monolithic baseline systems.

1.3 Contributions

This research makes three primary contributions to the literature on human-centred AI and design science:

1. **Theoretical Contribution:** We establish a theoretical foundation for decomposed AI decision-making that integrates principles from multi-agent systems, explainable AI, and human-computer interaction.
2. **Methodological Contribution:** We develop and validate a complete seven-step design method that operationalises these principles into actionable guidance for AI system designers, demonstrated through the EODD Studio platform.
3. **Empirical Contribution:** We provide rigorous empirical evidence, through heuristic development with human experts ($N = 5$), scaled feasibility assessment via LLMs-as-Experts ($N = 100$), and comparative user studies ($N = 16$), demonstrating the method's effectiveness and generalisability—including the critical finding that improved interpretability does not increase cognitive load.

2. Literature Review

2.1 Explainable AI and Interpretability

The challenge of AI interpretability has been a central concern in machine learning research, particularly as deep learning models have achieved superior performance at the cost of transparency (Lipton, 2018). Post-hoc explanation methods, such as LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017), attempt to explain model predictions by approximating local decision boundaries or attributing importance to input features. While these approaches have gained widespread adoption, they are fundamentally limited: they explain approximations of model behaviour rather than the actual reasoning process (Rudin, 2019).

More recent work has explored attention mechanisms as a source of interpretability (Vaswani et al., 2017), but research has demonstrated that attention weights do not reliably indicate which inputs influence model outputs (Jain & Wallace, 2019). Furthermore, Wiegreffe and Pinter (2019) showed that attention-based explanations can be misleading when models learn to manipulate attention patterns without changing their underlying predictions. These limitations motivate our approach of designing interpretability into the system architecture itself, as Kadir (2026) proposed in the ensemble of specialised LLMs architecture, where interpretable pedagogical orchestration replaces opaque single-model inference.

2.2 Multi-Agent Systems and LLM Orchestration

The emergence of LLMs with sophisticated reasoning capabilities has enabled a new paradigm of multi-agent systems where specialised agents collaborate on complex tasks (Wang et al., 2024; Xi et al., 2023). Research has demonstrated that decomposing complex problems into sub-tasks handled by specialised agents can improve both performance and interpretability (Wu et al., 2023). However, existing approaches typically focus on task decomposition for performance optimization rather than systematic design for interpretability and human oversight.

Recent frameworks such as AutoGPT (Significant Gravitas, 2023) and LangChain (Chase, 2023) provide infrastructure for LLM agent orchestration but do not prescribe design principles for interpretable decision-making. Similarly, agent communication protocols (Wooldridge & Jennings, 1995) have been extensively studied in the multi-agent systems literature, but their application to human-centred AI oversight remains underexplored. Our work bridges this gap by developing explicit orchestration rules that both coordinate agent behaviour and generate

human-auditable traces, building on the multi-agent conversation framework proposed by Wu et al. (2023).

2.3 Human-Centred AI and Oversight

The principle of meaningful human oversight has gained prominence in AI governance frameworks, including the EU AI Act (European Commission, 2021) and IEEE standards for ethically aligned design (IEEE, 2019). However, translating these principles into practical system designs remains challenging. Research on human-AI collaboration has demonstrated that effective oversight requires not just explanations but also mechanisms for humans to inspect, challenge, and override AI decisions (Amershi et al., 2019; Eslami et al., 2018).

The concept of 'human-in-the-loop' has been widely adopted, but implementations often relegate humans to passive oversight roles with limited ability to understand or intervene in AI decision processes (Green & Chen, 2019). Our work advances this literature by designing explicit human review checkpoints into the decision architecture, supported by comprehensive decision traces that enable meaningful inspection and intervention, consistent with the human-AI interaction guidelines proposed by Amershi et al. (2019).

2.4 Cognitive Load and AI Interaction

A critical but often overlooked aspect of AI system design is the cognitive burden placed on users. The NASA Task Load Index (NASA-TLX; Hart & Staveland, 1988) provides a validated framework for assessing mental demand, physical demand, temporal demand, performance, effort, and frustration. Research on human-AI interaction has shown that providing more information does not always improve user performance if it increases cognitive load beyond optimal levels (Paas & van Merriënboer, 1993).

In the context of explainable AI, there is an inherent tension: more detailed explanations may improve transparency but at the cost of increased mental effort. This creates a potential trade-off that system designers must navigate. Our research explicitly tests whether the EODD method's improvements in interpretability and trustworthiness come at the cost of increased cognitive load, addressing a critical gap in the literature on human-centred AI design. Following Paas and van Merriënboer (1993), we conceptualise cognitive load efficiency as the relationship between performance outcomes and invested mental effort.

2.5 Design Science Research in AI

Design Science Research (DSR) provides a rigorous methodological framework for creating and evaluating innovative artefacts (Hevner et al., 2004; Gregor & Hevner, 2013). In the context of AI systems, DSR has been applied to develop novel algorithms, system architectures, and design methods (Vom Brocke et al., 2020). The DSR cycle of problem identification, artefact design, evaluation, and communication provides a structured approach to developing practically relevant and scientifically rigorous contributions.

Our research follows the DSR methodology, treating EODD as a design artefact that addresses the identified problem of opaque AI decision systems. We adopt the evaluation framework proposed by Venable et al. (2016), combining expert evaluation for assessing design quality with empirical user studies for demonstrating practical utility. This dual-evaluation approach strengthens the validity of our findings and supports claims of generalisability.

3. Methodology

This research follows the Design Science Research (DSR) methodology (Hevner et al., 2004; Gregor & Hevner, 2013), which provides a rigorous framework for developing and evaluating innovative design artefacts. The DSR process comprises three interconnected cycles: the Relevance Cycle, which connects the research to the problem environment; the Design Cycle, which iteratively develops and refines the artefact; and the Rigor Cycle, which grounds the research in existing knowledge bases.

3.1 Design Cycle: EODD Method Development

The EODD method was developed through an iterative design cycle involving literature synthesis, expert consultation, and prototype refinement. The resulting seven-step method provides systematic guidance for architecting interpretable AI decision systems.

Step 1: Decision Context Definition

The first step establishes the boundaries and requirements for the decision system. Designers must identify the stakeholders affected by decisions, the consequences of errors, applicable regulatory requirements, and the human oversight model. This contextual grounding ensures that subsequent design decisions are aligned with real-world constraints and stakeholder needs.

Step 2: Decision Decomposition

Complex decisions are decomposed into constituent sub-decisions, each addressing a specific aspect of the overall problem. This decomposition follows principles of functional separation, where distinct reasoning tasks (e.g., evidence assessment, constraint checking, ethical evaluation) are identified and isolated. The decomposition enables specialised handling of each sub-decision and creates natural boundaries for generating interpretable traces.

Step 3: Specialised LLM Role Assignment

Each sub-decision is assigned to a specialised LLM agent with a defined role, responsibilities, and constraints. The EODD method prescribes four core roles: (1) the Reasoner, which performs primary evidence-based reasoning; (2) the Checker, which validates the Reasoner's conclusions against domain constraints; (3) the Ethics Reviewer, which evaluates decisions against ethical principles and stakeholder values; and (4) the Explainer, which synthesises the decision process into human-understandable explanations. Additional domain-specific roles may be defined as needed. This architecture draws on the ensemble approach described by Kadir (2026), where specialised LLM agents collaborate through interpretable orchestration.

Step 4: Orchestration Rules Specification

Orchestration rules define how agents interact, including sequencing constraints (which agents must complete before others begin), data flow specifications (what information passes between agents), and human trigger points (where human review is required). These rules are explicit, auditable, and can be validated independently of the agents themselves, providing a layer of transparency not available in monolithic systems.

Step 5: Decision Synthesis

The outputs of individual agents are synthesised into a final decision through a defined aggregation mechanism. This may involve simple majority voting, weighted combination based on agent confidence, or more sophisticated consensus-building algorithms. The synthesis process itself is recorded in the decision trace, enabling post-hoc analysis of how individual agent contributions influenced the final outcome.

Step 6: Decision Trace Generation

A comprehensive decision trace is generated for each decision, capturing: (a) the rationale log—each agent's reasoning and conclusions; (b) role contributions—what each agent contributed to the decision; (c) assumptions—explicit and implicit assumptions made during

reasoning; and (d) confidence scores—each agent's assessment of its conclusion reliability. This trace serves as the primary mechanism for human oversight and post-hoc audit.

Step 7: Human Review Integration

Human review capabilities are integrated at defined checkpoints throughout the decision process. These checkpoints enable humans to inspect the decision trace, challenge specific agent conclusions, and override decisions when necessary. The design of review interfaces must balance comprehensiveness with usability, presenting the decision trace in a format that supports rapid assessment while preserving detail for deep investigation.

3.2 Evaluation Methodology

The framework was evaluated through a three-phase methodology designed to assess theoretical validity, implementation feasibility, and empirical user outcomes.

Phase 1: Heuristic Development (N=5 Human Experts)

The first phase involved five specialized domain experts: a Chief Medical Information Officer (CMIO), an AI in Education Researcher, a Senior AV Safety Validation Engineer, and a Visual Communication/HCD Researcher. The evaluation method utilized focus groups and semi-structured, interactive interviews to critically evaluate the theoretical framework.

Phase 2: Scaled Feasibility Assessment (N=100 LLMs-as-Experts)

To build upon the initial expert feedback, the second phase deployed an innovative "LLMs-as-Experts" framework. This simulated 100 specialized domain personas to conduct a scaled heuristic evaluation of the artifact. The primary objective of this phase was to quantify broad implementation viability and anticipate potential areas of friction prior to human testing.

Phase 3: Empirical User Validation (N=16)

The final phase was a within-subjects, counterbalanced comparative study involving 16 participants, which included graduate-level peers, Teaching Assistants (TAs), and domain novices.

1. **Design & Systems:** Participants evaluated an explicitly orchestrated EODD intervention (System Y) against opaque Monolithic Baselines (System X). The

baselines included Tesla Model S 2021–2026 (Autonomous Vehicles), Rokid AI Smart Glasses 2025 (AR/VR), and ChatGPT 3.5 2022 (Learning Systems).

2. **Test Scenarios:** The study utilized 15 high-stakes scenarios across 3 distinct domains (Autonomous Vehicles, AR/VR and Learning Systems). These scenarios incorporated 5 specific decision archetypes: Generic Interaction, Error Handling, Assisted AI Interaction, Information Retrieval, and Sustained Enquiry.
3. **Quantitative Metrics:** Participants rated Interpretability, Trustworthiness, and Cognitive Load (using NASA-TLX) via 7-point Likert scales. Additionally, Trace Quality and Error-Spotting Accuracy were measured using objective 0–7 scoring rubrics.
4. **Qualitative Analysis:** Think-aloud protocols and post-task semi-structured interviews were conducted and subsequently analyzed using thematic coding.

3.3 Data Analysis

Quantitative data were analysed using paired-samples t-tests to compare System X (monolithic) and System Y (EODD) ratings. Effect sizes were calculated using Cohen's d for paired samples. Assumptions of normality were verified through visual inspection of Q-Q plots and Shapiro-Wilk tests. All analyses were conducted using Python (SciPy v1.11) with significance level set at $\alpha = .05$. Qualitative data from think-aloud protocols were analysed using reflexive thematic analysis (Braun & Clarke, 2021). Inter-rater reliability for LLM-vs-human comparisons was assessed using Cohen's kappa.

4. Results

4.1 Phase 1 & Phase 2: Heuristic Development and Scaled Feasibility Results The evaluation across Phase 1 (N = 5 human experts) and Phase 2 (N = 100 simulated LLMs-as-Experts) demonstrated strong support for the EODD method's engineering feasibility and overall promise. Qualitative insights from the focus groups and semi-structured interviews with the five specialized human experts in Phase 1 informed the baseline understanding of the theoretical framework's strengths and implementation challenges.

Metric	Scale	Mean (SD)	% Rating Favourably
Engineering Feasibility	1–7	5.66 (0.83)	92.0% (≥ 5)
Overall Promise	1–5	4.51 (0.50)	98.0% (≥ 4)
No Domain Friction	Binary	47%	47% reported no friction

Table 1: Scaled Feasibility Assessment Results ($N = 100$ LLMs-as-Experts)

To quantify broad implementation viability, the Phase 2 scaled feasibility assessment utilizing 100 specialized domain personas produced the metrics summarized in Table 1. Engineering Feasibility received a mean rating of 5.66 ($SD = 0.83$) on a 7-point scale, with 92% of the simulated experts rating it 5 or above. Overall Promise received a mean rating of 4.51 ($SD = 0.50$) on a 5-point scale, with 98% rating it 4 or above. Regarding domain generalizability, 47% of the LLM personas reported no significant friction in applying the method to their specific domain. Among the simulated personas that anticipated friction, the primary concerns were real-time/latency constraints (28%), difficulty defining appropriate orchestration rules (19%), and lack of standardizable roles (6%). Analysis across the simulated professional domains (Autonomous Vehicles, Healthcare AI, Financial Systems, Legal AI, and Educational Technology) revealed consistent patterns of support aligned with the initial human expert feedback.

4.2 Comparative User Study Results

The within-subjects comparative study ($N = 16$) provided strong empirical support for all three hypotheses. Table 2 presents the complete statistical results including mental demand.

Metric	System X (Mono.)	System Y (EODD)	t(15)	p-value	Cohen's d
Interpretability (H1)	3.29 (1.16)	5.28 (1.27)	6.42	< .001***	1.61
Trustworthiness (H2)	3.15 (1.15)	5.10 (1.30)	5.87	< .001***	1.47
Error Confidence (H3)	3.55 (1.37)	5.14 (1.25)	4.71	< .001***	1.18
Trace Quality	2.48 (1.11)	5.11 (1.40)	7.23	< .001***	1.81
Mental Demand	4.42 (1.50)	3.82 (1.43)	−0.93	.37 (ns)	−0.24

Table 2: Comparative Study Results ($N = 16$, paired-samples *t*-tests)

Note. *M* = Mean; *SD* = Standard Deviation. All ratings on 1–7 Likert scales. *** $p < .001$; ns = not significant.

4.2.1 Hypothesis 1: Interpretability

EODD-based systems ($M = 5.28$, $SD = 1.27$) received significantly higher interpretability ratings than monolithic systems ($M = 3.29$, $SD = 1.16$), $t(15) = 6.42$, $p < .001$, Cohen's $d = 1.61$. This large effect size indicates that the decomposed, trace-based architecture of EODD substantially improves users' ability to understand AI decisions. The mean difference of 1.99 points on a 7-point scale represents a practically significant improvement in interpretability.

4.2.2 Hypothesis 2: Trustworthiness

EODD-based systems ($M = 5.10$, $SD = 1.30$) received significantly higher trustworthiness ratings than monolithic systems ($M = 3.15$, $SD = 1.15$), $t(15) = 5.87$, $p < .001$, Cohen's $d = 1.47$. This large effect size demonstrates that the explicit reasoning traces and multi-agent validation in EODD substantially enhance user trust in AI systems.

4.2.3 Hypothesis 3: Oversight Performance

Users reported significantly higher confidence in identifying errors within EODD-based systems ($M = 5.14$, $SD = 1.25$) compared to monolithic systems ($M = 3.55$, $SD = 1.37$), $t(15) = 4.71$, $p < .001$, Cohen's $d = 1.18$. This large effect size indicates that the decision traces generated by EODD enable more effective human oversight by making potential errors more detectable.

4.2.4 System Preference

Across all 240 scenario evaluations (16 participants x 15 scenarios), participants preferred the EODD-based system in 76.2% of cases (183/240), while preferring the monolithic system in only 23.8% of cases (57/240). This strong preference for EODD aligns with the quantitative ratings and suggests that the benefits of decomposed reasoning and explicit traces are readily apparent to users.

4.2.5 Trace Quality Analysis

The composite trace quality score (averaging clarity, completeness, transparency, and actionability) showed the largest effect size: EODD ($M = 5.11$, $SD = 1.40$) versus monolithic ($M = 2.48$, $SD = 1.11$), $t(15) = 7.23$, $p < .001$, Cohen's $d = 1.81$. This very large effect size underscores that the explicit decision traces generated by EODD are substantially more useful for human oversight than the opaque outputs of monolithic systems.

4.2.6 Mental Demand Analysis

A critical concern in designing interpretable AI systems is whether enhanced transparency comes at the cost of increased cognitive load. To address this question, we measured mental demand using the NASA-TLX mental demand subscale (1 = Very low effort, 7 = Very high effort).

Results revealed no significant difference in mental demand between System X (monolithic: $M = 4.42$, $SD = 1.50$) and System Y (EODD: $M = 3.82$, $SD = 1.43$), $t(15) = -0.93$, $p = .37$, Cohen's $d = -0.24$. Notably, the EODD system showed a trend toward lower mental demand (0.6 points lower on the 7-point scale), though this difference did not reach statistical significance.

This finding carries important practical implications: the substantial improvements in interpretability, trustworthiness, and error detection achieved by EODD do not come at the cost of increased cognitive burden. Users can benefit from enhanced transparency without experiencing greater mental effort, suggesting that well-designed decision traces can make complex AI reasoning more accessible without overwhelming users.

The non-significant trend toward lower mental demand for EODD may reflect the structured nature of the decision trace, which presents information in a predictable, organised format. In contrast, participants evaluating monolithic systems may have experienced greater cognitive effort attempting to infer the reasoning process from limited output. This interpretation aligns with think-aloud protocol findings, where participants frequently expressed frustration with the 'guesswork' required to understand monolithic decisions.

4.3 Qualitative Findings

Thematic analysis of think-aloud protocols revealed consistent patterns in how participants engaged with the two systems. Four primary themes emerged:

Theme 1: Structural Transparency Enables Deeper Inspection. Participants consistently reported that the decomposed structure of EODD traces enabled them to pinpoint specific reasoning steps for closer examination. Representative quote: 'With System Y, I can see exactly where the reasoning went wrong—it's in the ethics check step.'

Theme 2: Role Separation Builds Confidence. The presence of multiple agents checking each other's work was cited as a source of confidence. Representative quote: 'Having a separate checker agent makes me feel like the answer has been validated, not just generated.'

Theme 3: Monolithic Opacity Hinders Error Detection. Participants frequently expressed frustration with the monolithic system's lack of transparency. Representative quote: 'System X just gives me an answer—I have no idea how it got there, so I can't tell if it's right.'

Theme 4: Explicit Assumptions Support Critical Evaluation. The explicit listing of assumptions in EODD traces was valued for enabling critical evaluation. Representative quote: 'When I can see what assumptions the system made, I can judge whether they're valid for this case.'

4.4 Triangulation Analysis

Triangulation across the three evaluation phases strengthens confidence in the findings. Expert evaluation established design feasibility and domain generalisability; comparative user study demonstrated superior performance on all hypothesised metrics without increased cognitive load; qualitative analysis revealed mechanisms underlying these quantitative advantages; and LLMs-as-Experts validation confirmed the scalability of findings. The convergence of findings across methods and participant populations supports the validity of conclusions regarding EODD's effectiveness.

5. Discussion

5.1 Theoretical Implications

This research makes several contributions to theory in human-centred AI and explainable systems. First, we demonstrate that architectural decomposition—structuring AI decisions as orchestrated multi-agent processes—can achieve interpretability by design rather than through post-hoc explanation. This finding challenges the prevailing assumption that interpretability must be traded against performance, showing instead that decomposed architectures can enhance both, as proposed by Kadir (2026) in the ensemble LLM architecture for adaptive tutoring.

Second, our results support the theoretical proposition that explicit decision traces serve as a critical mediating artefact for human oversight (Amershi et al., 2019). The very large effect size for trace quality ($d = 1.81$) indicates that the format and content of decision explanations

profoundly influence oversight effectiveness. This suggests that future research on explainable AI should focus not merely on generating explanations but on designing trace structures that support specific oversight tasks.

Third, the finding that 47% of experts anticipated no domain friction suggests that the EODD method captures domain-independent design principles. This aligns with the DSR goal of producing generalisable knowledge (Gregor & Hevner, 2013) and positions EODD as a kernel theory that can be instantiated across diverse application contexts.

Fourth, and perhaps most importantly, our finding that enhanced interpretability does not increase mental demand challenges a common assumption in the explainable AI literature. Previous work has often assumed a trade-off between explanation richness and cognitive load (Langer et al., 2021). Our results suggest that well-designed decision traces can present complex information in ways that are cognitively manageable, potentially through structured presentation and progressive disclosure, consistent with the instructional efficiency framework of Paas and van Merriënboer (1993).

5.2 Practical Implications

For practitioners designing AI decision systems, this research provides actionable guidance. The seven-step EODD method offers a systematic process for architecting interpretability into systems from the outset, rather than attempting to retrofit explanations onto opaque models. The strong expert ratings for engineering feasibility ($M = 5.66/7$) suggest that the method is practically implementable with existing LLM and orchestration technologies.

The identification of latency concerns as a primary friction point (28% of experts) highlights a practical constraint that designers must address. For real-time applications, optimisations such as parallel agent execution, caching of intermediate results, or tiered decision traces (summary vs. detailed) may be necessary. These adaptations preserve the core EODD principles while addressing domain-specific constraints.

The mental demand findings have particular practical significance. System designers can implement EODD-based decision traces confident that enhanced transparency will not overwhelm users with additional cognitive burden. This removes a potential barrier to adoption that might otherwise arise from concerns about user workload.

For organisations deploying AI in regulated industries, EODD provides a structured approach to meeting emerging requirements for AI transparency and human oversight (European

Commission, 2021). The explicit orchestration rules and comprehensive decision traces can serve as audit artefacts, demonstrating compliance with regulatory expectations for explainability and meaningful human control.

The EODD Studio platform (<https://eodd.studio>) demonstrates the practical implementation of these principles, providing researchers and practitioners with a deployable reference architecture for orchestrated, interpretable AI decision systems.

5.3 Limitations and Future Research

Several limitations should be acknowledged. First, the comparative user study employed a scenario-based design rather than interaction with live systems. While this approach enabled controlled comparison across multiple domains, future research should validate findings with deployed systems in real-world contexts.

Second, the study focused on decision scenarios where ground truth could be established. The effectiveness of EODD in more ambiguous contexts—where even expert consensus is lacking—represents an important avenue for future investigation. Similarly, the method's applicability to non-textual decision contexts (e.g., computer vision, sensor fusion) requires further study.

Third, while the human expert sample in Phase 1 ($N = 5$) provided deep qualitative insights, it was small and convenience-sampled from professional networks. Future research should employ stratified sampling with a larger human cohort to ensure representation across different industry sectors, organization sizes, and geographic regions. Additionally, a notable limitation is that the Phase 2 scaled feasibility assessment relied on 100 simulated LLM personas (LLMs-as-Experts) rather than real human professionals. While this approach enabled rapid, large-scale heuristic evaluation, simulated judgments may not perfectly capture the nuanced, real-world operational frictions experienced by actual practitioners.

Fourth, the mental demand measure was limited to a single NASA-TLX subscale. Future research should employ the complete NASA-TLX (Hart & Staveland, 1988) to assess temporal demand, performance, effort, and frustration, providing a more comprehensive picture of cognitive load. Additionally, physiological measures such as eye-tracking or pupillometry could provide objective indicators of cognitive effort.

Future research directions include: (1) developing automated tools for generating orchestration rules from decision requirements; (2) investigating optimal trace visualisations for different

oversight tasks; (3) exploring adaptive orchestration where agent composition changes based on decision characteristics; (4) longitudinal studies of EODD system deployment in production environments; (5) investigating individual differences in cognitive load responses to different trace formats; and (6) scaling LLMs-as-Experts validation to broader domain coverage and cross-model comparison.

6. Conclusion

This paper presents Explicit Orchestrated Decision Design (EODD), a Design Science Research method for creating human-centred, interpretable AI decision systems. Through rigorous multi-phase evaluation, we demonstrate that EODD significantly outperforms monolithic baselines on interpretability, trustworthiness, and oversight performance, with large effect sizes indicating practical as well as statistical significance.

A particularly important finding is that these substantial improvements were achieved without significantly increasing mental demand on users. This challenges the assumption that enhanced transparency necessarily comes at the cost of increased cognitive burden, demonstrating that well-designed decision traces can make complex AI reasoning more accessible without overwhelming users.

The EODD method addresses a critical gap in current AI system design: the lack of systematic approaches for architecting interpretability and human oversight into decision systems. By structuring AI decisions as orchestrated multi-agent processes with explicit traces, EODD enables the kind of meaningful human oversight increasingly demanded by regulators, stakeholders, and society. The method's domain-agnostic design and strong expert ratings for feasibility position it as a practical tool for AI practitioners across diverse high-stakes domains.

The EODD Studio web application (<https://eodd.studio>) provides a tangible demonstration of these principles, offering researchers and practitioners an accessible platform for experimenting with orchestrated AI decision architectures.

As AI systems assume greater decision-making authority in consequential contexts, the principles embodied in EODD—decomposition, explicit orchestration, comprehensive tracing, and integrated human review—will become increasingly essential. This research contributes both a validated design method and empirical evidence supporting the broader proposition that interpretable AI is achievable not despite but through thoughtful architectural design, and that

such interpretability need not impose additional cognitive burden on the humans who must ultimately oversee these systems.

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Appendix A: EODD Method Detailed Specification

A.1 Specialised Agent Roles

The EODD method defines four core agent roles that should be instantiated in every decision system:

Reasoner: The primary reasoning agent responsible for evidence analysis and initial conclusion generation. The Reasoner processes available evidence, applies domain knowledge, and produces a preliminary decision with supporting rationale. Output: decision recommendation, confidence score, rationale log.

Checker: A validation agent that verifies the Reasoner's conclusions against domain constraints, logical consistency, and factual accuracy. The Checker identifies potential errors, contradictions, or omissions in the Reasoner's analysis. Output: validation report, identified issues, correction suggestions.

Ethics Reviewer: An evaluation agent that assesses decisions against ethical principles, stakeholder values, and potential harms. The Ethics Reviewer ensures decisions align with organisational values and societal expectations. Output: ethical assessment, identified concerns, mitigation recommendations.

Explainer: A synthesis agent that transforms the multi-agent reasoning process into human-understandable explanations. The Explainer aggregates role contributions and generates accessible decision traces. Output: decision trace, summary explanation, detailed rationale.

A.2 Orchestration Rule Types

Orchestration rules in EODD are specified using three primary constructs:

Sequencing Rules: Define the order in which agents must execute. Example: 'Checker must complete validation before Ethics Reviewer begins assessment.'

Data Flow Rules: Specify what information passes between agents. Example: 'Reasoner outputs (decision, confidence, rationale) are inputs to Checker.'

Human Trigger Rules: Define conditions requiring human review. Example: 'Human review required when Ethics Reviewer identifies potential harm OR Checker finds validation errors.'

A.3 Decision Trace Structure

The EODD decision trace comprises four mandatory fields:

Rationale Log: A chronological record of each agent's reasoning steps, including inputs considered, inferences made, and conclusions reached.

Role Contributions: A summary of what each agent contributed to the final decision, enabling assessment of individual agent influence.

Assumptions: An explicit listing of all assumptions made during the decision process, both by individual agents and through orchestration.

Confidence Scores: Each agent's self-assessed confidence in its contribution, enabling identification of low-confidence reasoning that may warrant additional scrutiny.

A.4 Mental Demand Measurement

Mental demand was measured using the NASA-TLX mental demand subscale (Hart & Staveland, 1988). Participants rated: 'How mentally demanding was the task?' on a 7-point scale where 1 = Very low effort, 4 = Moderate effort, and 7 = Very high effort. This single-item measure was selected for its established validity and minimal additional burden on participants.

A.5 EODD Studio Platform

The EODD Studio (<https://eodd.studio>) is a web-based implementation of the EODD method that enables users to design, configure, and deploy orchestrated AI decision systems. The platform provides visual tools for defining agent roles, specifying orchestration rules, generating decision traces, and integrating human review checkpoints. It serves as both a reference implementation for researchers and a practical tool for practitioners seeking to adopt the EODD method in production systems.

